

PRESSURIZED Process Documentation

Origins: Capstone I

PRESSURIZED began as a project called Control Deck during Capstone I. The initial prototype established the core concept: a Unity video game played not with a conventional controller, but with a handcrafted panel of physical inputs. The prototype featured a timer bar, basic command prompts displayed through hand-drawn images of inputs, and a simple scrolling underwater scene. While early user testing confirmed the concept was intriguing and that players always wanted to try again, it also revealed critical weaknesses: command prompts were initially too vague to parse quickly, and after extended play, the game became repetitive and too easy. Users often ignored the submarine visuals entirely, undermining the visual design.

Capstone II: Planning and Scope Definition

At the start of Capstone II, the project entered an analysis phase. Four milestones were defined for the semester:

1. Increase Interactive Game Elements: Add a specimen collection mechanic, boss battles, and dynamic creature spawning to deepen gameplay.
2. Cater to the Arcade Format: Implement a complete game loop with a start screen, end-game scoring screen, and combo-based dynamic scoring.
3. Reformat and Expand the Controller: Replace image-based command prompts with text and symbol labels; expand the board significantly; add new input types.
4. Develop an Arcade Cabinet: Construct a physical arcade cabinet as a stretch goal, unifying the project into a single presentable unit.

Early Semester: Ideation and Code Refactoring

The early weeks of the semester were spent revisiting and cleaning the Capstone I codebase, which had been developed rapidly without consideration for future expansion. Scripts were refactored to support modularity. The biome system was overhauled from two static biomes to a randomized system capable of transitioning between multiple distinct

underwater environments. Sketches were produced for creature animations, the grid-based boss mechanic, and the end-game scoring screen.

Milestone 2: Game Systems Completed

By mid-semester, the core game systems were fully implemented. The boss battle system was built: creatures spawn at a score threshold and move between screen zones correlated with physical buttons on the controller. The scoring system was updated to reward faster reactions with higher point multipliers. Multiple biomes were scripted with distinct boss types and fully customizable sprite and movement variables. An official title, PRESSURIZED, was selected and a logo and start menu animation were created. A gameplay video was produced as documentation at this stage.

Physical Development Timeline

March 21: Logo created.

March 23: New physical inputs sourced and components prepared.

April 2: Arcade cabinet CAD drawings finalized in CAD software with real-world dimensions and ADA-compliant control heights.

April 4–11: Cabinet construction: four 4x8 plywood sheets and four 2x4 studs cut, sanded, and assembled. Cabinet construction finalized April 11.

April 6: Initial controller layout laid out and photographed.

April 9: Early wiring experiments and input testing.

April 10: Paper prototyping of final controller layout on cardboard at full scale.

April 14: PRESSURIZED showcase poster created.

April 18: Full crunch phase begins. Controller and cabinet in development.

April 19: Controller CAD drawings finalized.

April 21: Control panel design finalized and sent for laser cutting.

April 22: Control panel laser cut and etched with module labels.

April 26: Cardboard module dividers fabricated and installed inside the controller housing to organize wiring.

April 30: Initial input wiring completed across all modules.

May 1: LED backlighting strip installed; tertiary Arduino Uno added to handle lighting data.

May 3: Full unit debugging and lighting tests conducted. All systems run together for first time.

May 4: Marquee created and installed above the cabinet screen.

May 6: Cabinet relocated to presentation venue.

May 7: Presentation rehearsals and final playtesting with peers.

May 8: Final showcase presentation.

Key Decisions and Pivots

One significant pivot occurred when the original plan for a fully enclosed, seated arcade booth was reconsidered. While the immersive booth concept was a long-held vision, it was determined to be too labor-intensive and would limit audience viewing. The decision was made to build a classic upright arcade cabinet instead, which provided equivalent craft and polish while remaining more accessible and viewable to a crowd. This decision proved productive: the cabinet was completed ahead of schedule, freeing time for controller development.

Another pivot involved command prompt design. Early prototypes used hand-drawn image prompts to identify inputs. User testing showed these were too slow to parse. The final design replaced images with text labels which were laser-etched onto the panel, creating faster readability and a stronger narrative connection between the on-screen events and physical inputs.

A final, last-minute change occurred in the board's backlighting. Update intervals sent to the LED strip occasionally cause an unpleasant flashing, distracting the player. Rather than using individual module lighting, a solid background light was used instead for presentations.

Technical Challenges

The most persistent technical challenge was the serial communication system between Unity and the Arduino-based controller. The controller maxed out the RAM on two Arduino Mega boards and an Arduino Uno, requiring all processing logic to remain in Unity rather than on the microcontrollers. Input states for some modules were initially inverted due to code written before the physical hardware existed, requiring variable flips throughout the command manager script. LED animation data had to be sent from Unity to the Arduino rather than processed locally, requiring reworked serial communication scripts.